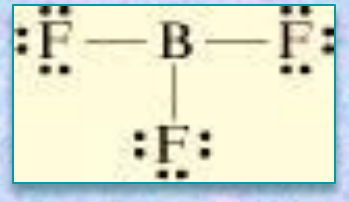
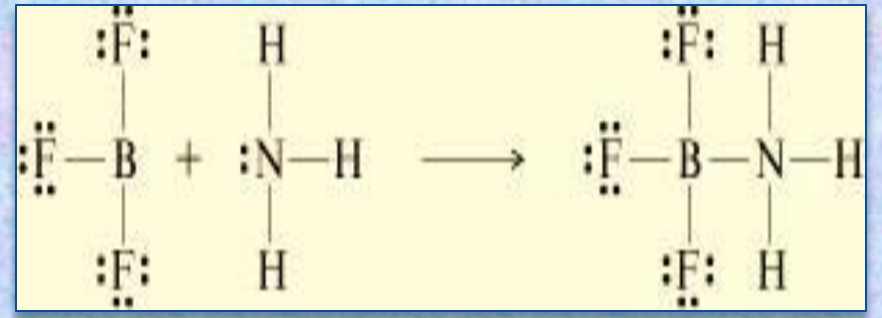
Electron Deficient Compound

Electron-deficient compounds are compounds in which an element has an incomplete octet.

H, Be, and B, often have fewer than eight electrons in Lewis structures.

For example, BF_3 is an electron-deficient compound.





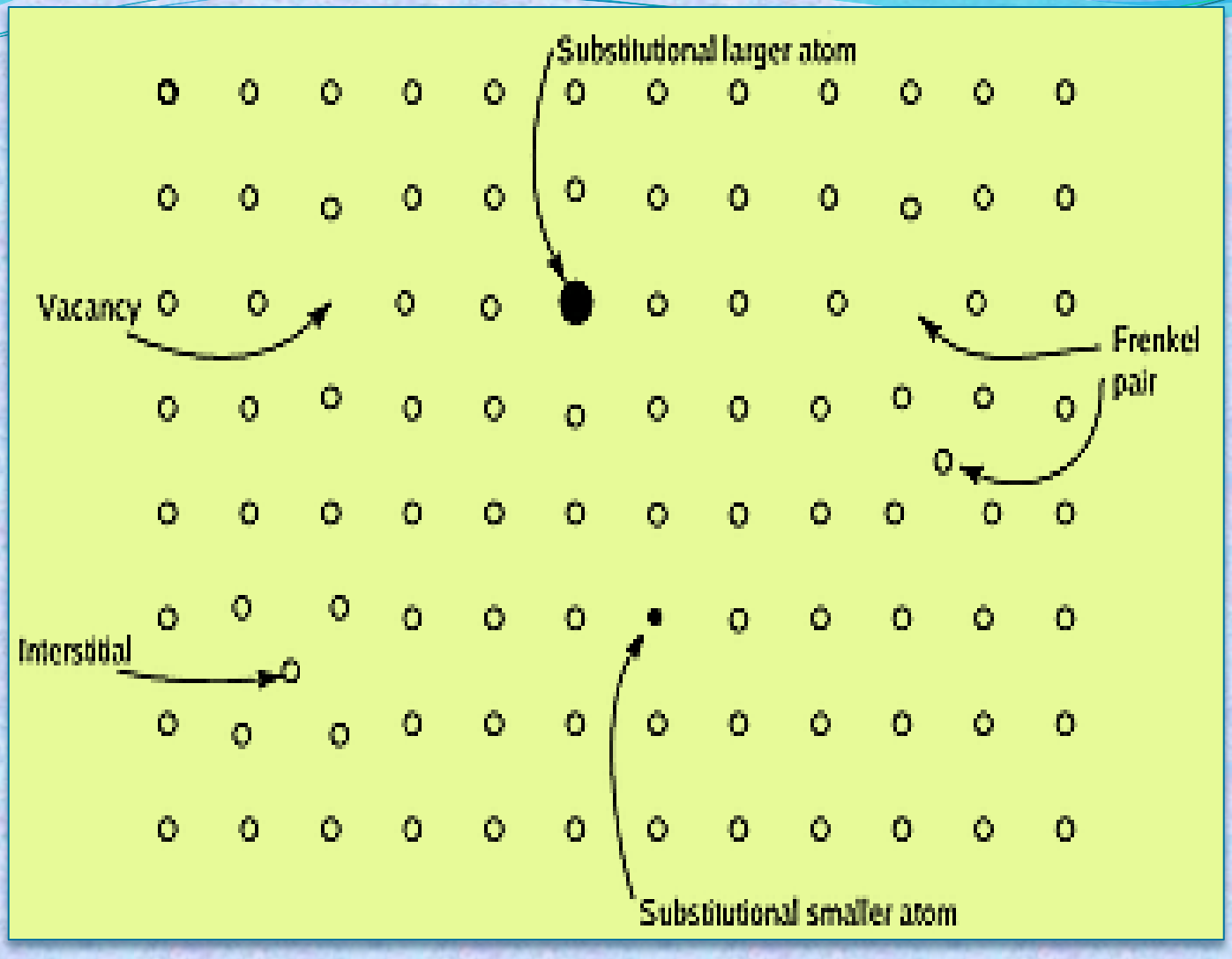
Non-stoichiometric Compounds

- **Non-stoichiometric compounds are chemical compounds with an elemental composition that cannot be represented by a ratio of *well-defined natural numbers*, and therefore violate the *law of definite proportions.***

- **Non-stoichiometric compounds are also known as *berthollides* (as opposed to the stoichiometric compounds or *daltonides*).**

Example

- **Although wüstite (ferrous oxide) has an ideal (stoichiometric) formula FeO, the actual stoichiometry is closer to $Fe_{0.95}O$.**
- **The non-stoichiometry occurs because of the ease of oxidation of Fe^{2+} to Fe^{3+}**
- **Thus for every three "missing" Fe^{2+} ions, the crystal contains two Fe^{3+} ions to balance the charge.**
- **Thus, the formula for wüstite is written as $Fe_{1-x}O$, where *x* is a small number (0.05 in the previous example) representing the deviation** from the "ideal" formula.



Crystallographic defects can cause solids to be non-stoichiometric

Boron Hydride or Borane

Alfred Stock established borane Chemistry

The compounds of Boron with Hydrogen is called Boron hydride or Borane.

B + H = Borane

Two Classes:

1. $B_nH_{(n+4)-}$ **Nido - borane**
2. $B_nH_{(n+6)-}$ **Arachno -borane**

Examples

Nido

B_2H_6 - Diborane
B_5H_9 - Pentaborane -9
B_6H_{10} - Hexaborane-10
B_8H_{12} - Octaborane- 12
$B_{10}H_{14}$ - Decaborane

$B_2H_6 \rightleftharpoons 2BH_3$
$BH_3 + B_2H_6 \rightarrow B_3H_7 + H_2$ (rate determining step)
$BH_3 + B_3H_7 \rightleftharpoons B_4H_{10}$
$B_2H_6 + B_3H_7 \rightarrow BH_3 + B_4H_{10}$
$\rightleftharpoons B_5H_{11} + H_2$

Arachno

B_4H_{10} - Tetraborane
B_5H_{11} – Pentaborane -11
B_6H_{12}- Hexaborane-12
B_8H_{14} –Octaborane-14
B_9H_{15}- Nonaborane

B_2H_6, is a colorless gas at room temperature. It is believed to be in equilibrium with borane, BH_3, according to the equation,

$$2BH_3 \rightleftharpoons B_2H_6$$

The Hydride of Boron/ Diborane (B_2H_6)

- **Synthesis:**
- $4BCl_3 + 3LiAlH_4 \rightarrow 2B_2H_6 + 3LiAlCl_4$
- $4BF3 + 3NaBH_4 \rightarrow 2\ B_2H_6 + 3NaBF_4$
- $2NaBH_4 + I_2 \rightarrow 2NaI + B_2H_6 + H_2$
- $8\ BF_3 + 6LiH \rightarrow B_2H_6 + 6LiBF_4$
- $BF_3 + 6NaH \longrightarrow B_2H_6 + 6NaF$

(Industrial Method)

Reaction of B_2H_6

1. With oxygen (explosive):

$$B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$$

2. With water:

$$B_2H_6 + 3H_2O \rightarrow B(OH)_3 + 6H_2$$

or alcohols:

$$B_2H_6 + 3HOR \rightarrow B(OR)_3 + 6H_2$$

3. Substitution reactions:

$$B_2H_6 + HCl \rightarrow B_2H_5Cl + H_2$$
$$B_2H_6 + 6Cl_2 \rightarrow 2BCl_3 + 6HCl$$

4. Cleavage with Lewis bases:

Symmetrical:

$$B_2H_6 + N(CH_3)_3 \rightarrow 2H_3BN(CH_3)_3$$

Unsymmetrical:

$$B_2H_6 + 2NH_3 \rightarrow [H_2B(NH_3)_2]^+[BH_4]^-$$

5. Reduction:

$$2B_2H_6 + 2Na \rightarrow NaBH_4 + NaB_3H_8$$
$$B_2H_6 + NaBH_4 \rightarrow NaB_3H_8 + H_2$$
$$5B_2H_6 + 2NaBH_4 \rightarrow Na_2B_{12}H_{12}$$

Addition of a H-B bond to C-C double bonds- i.e hydroboration reaction- alkylborane

$$1/2\ B_2H_6 + 3RCH{=}CHR \longrightarrow B(CH_2\text{-}CH_2R)_3$$

$$1/2\ B_2H_6 + 3RC{\equiv}CR \longrightarrow B(RC{=}CHR)_3$$

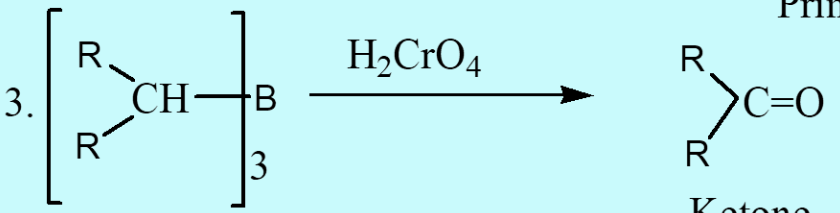
Alkylboran-BR_3

1. $BR_3 + 3CH_3COOH \longrightarrow 3RH + B(CH_3COO)_3$

Hydrocarbon

2. $B(CH_2\text{-}CH_2R)_3 + H_2O_2 \longrightarrow 3RCH_2CH_2OH + H_3BO_3$

Primary alcohol

3. 

Ketone

4. $(CH_3\text{-}CH_3){-}B \xrightarrow{H_2CrO_4} CH_3COOH$

Carboxylic acid

Reaction with Amonia

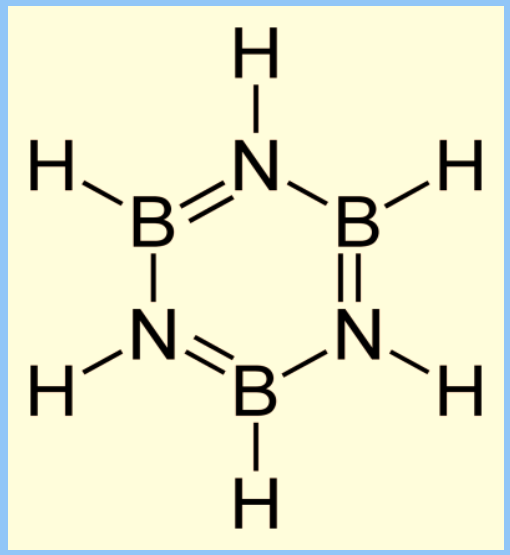
$$B_2H_6 + NH_3 \xrightarrow[\text{Low Temp.}]{\text{Excess}} B_2H_6{:}2NH_3 \quad \text{Ionic}$$

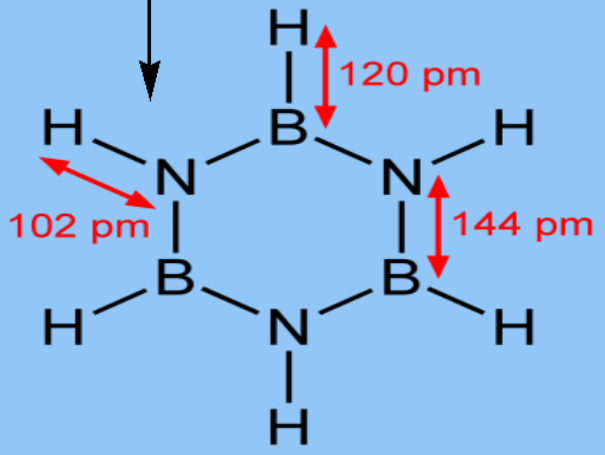
$$B_2H_6 + NH_3 \xrightarrow[\text{High Temp.}]{\text{Excess}} (BN)x \quad \text{Boron Nitride}$$

$$B_2H_6 + 2NH_3 \xrightarrow{\text{High Temp.}} B_3N_3H_6 \quad \text{Borazine}$$

$B_2H_6:2NH_3$ is a ionic compound exists as $[H_3N \rightarrow BH_2 \leftarrow NH_3]^+[BH_4]^-$

Temp.





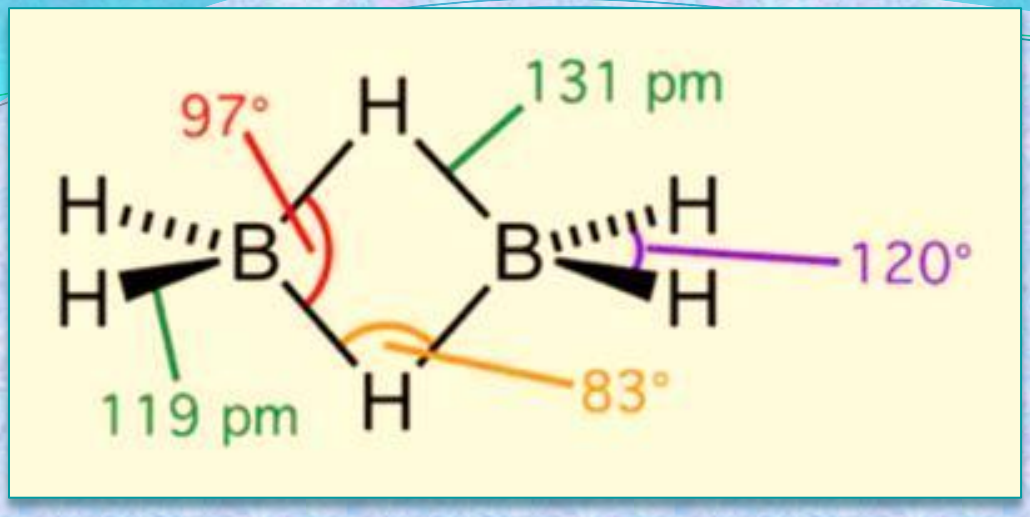
Borazine

Bonding in Diborane, B_2H_6

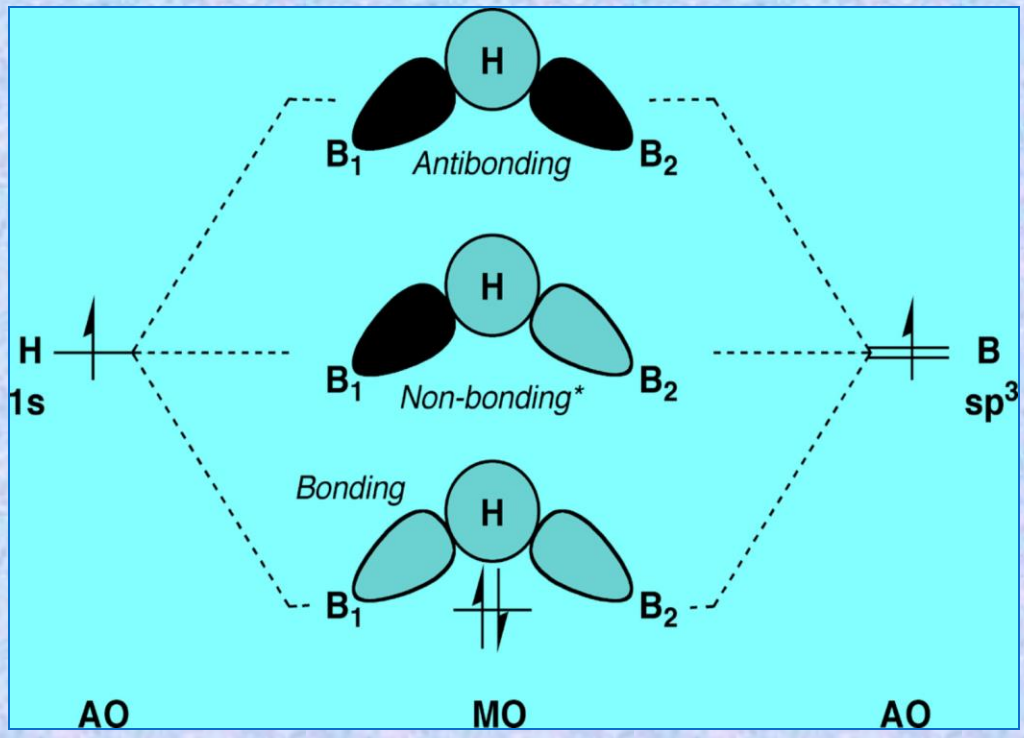
B(5) (ground)	1s2	2s2	$2p_x1$	$2p_y^0$	$2p_z^0$
(excited)	1s2	2s1	$2p_x1$	$2p_y1$	$2p_z^0$

Sp3- hybridization

The **hybridization** B(5) is sp^3



According to VBT



According to MOT

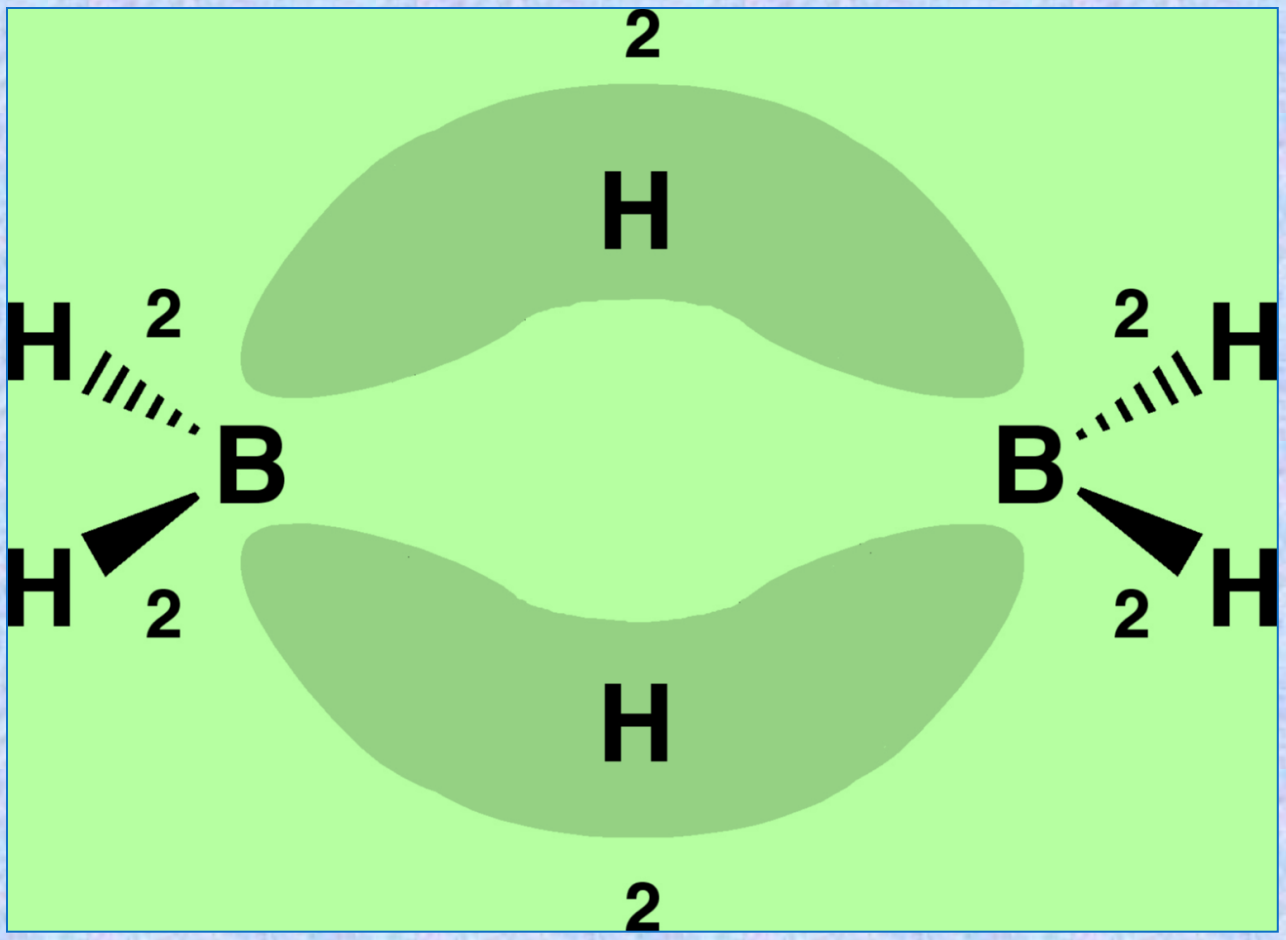
Three-center two-electron bond

A **three-center two-electron bond** is an electron deficient chemical bond where ***three atoms share two electrons.***

- The combination of three atomic orbitals form three molecular orbitals: ***one bonding, one non-bonding, and one anti-bonding.***
- The two electrons go into the bonding orbital, resulting in a net bonding effect and constituting a chemical bond among all three atoms.
- In many common bonds of this type, the bonding orbital is shifted towards two of the three atoms instead of being spread equally among all three.

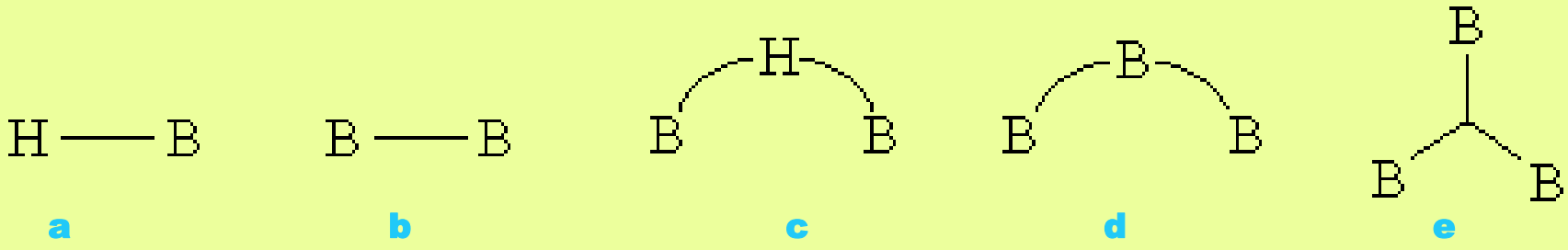
- When visualised in 3D, it looks like a banana, and hence a *3-center-2-electron bond* is often called a *banana bond*.

The MO scheme for one of the B–H–B bridging three center two electron bonds. *This picture is still a simplification of the actual MO scheme. The non-bonding orbital is actually of slightly lower energy than shown and so has slight bonding character. This arises from the fact that the orbitals involved in the terminal B–H bonding have the correct symmetry to overlap with the bridging bond orbitals, resulting in a stabilization of the ‘nonbonding’ orbital.



Banana Bond

Five Bonding Elements in Higher Borane



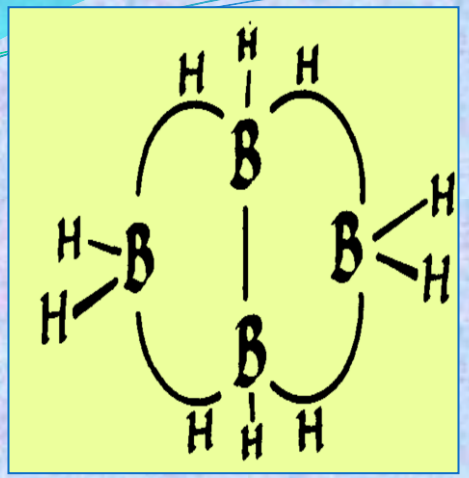
(a) Terminal 2c-2e Boron-Hydrogen bond

(b) 2c-2e boron-born bond

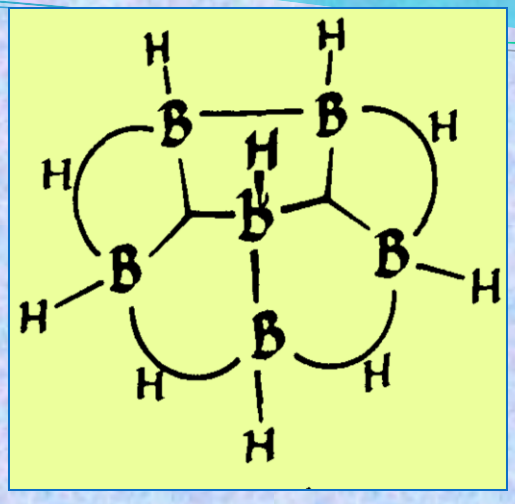
(c) 3c-2e boron-hydrogen-boron bridge bond

(d) Closed 3c-2e boron-boron-boron bond

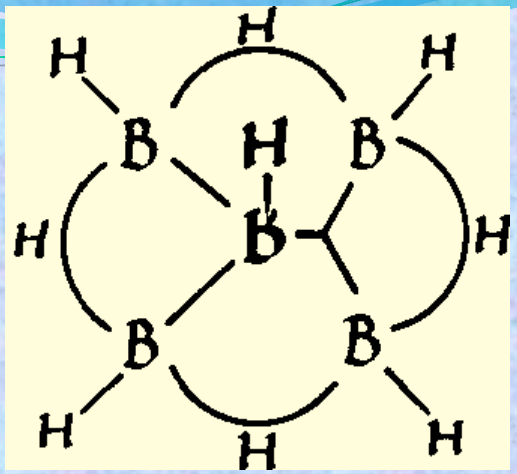
(e) Open 3c-2e boron-boron-boron bond



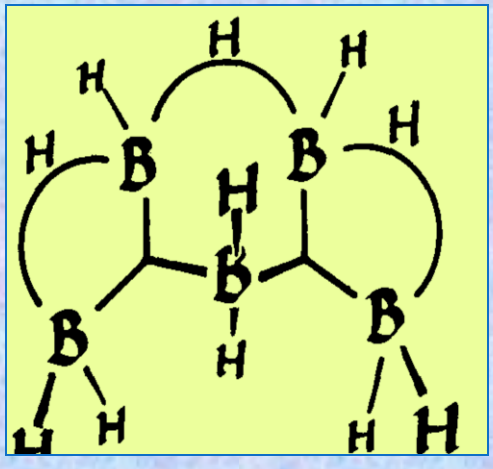
B_4H_{10}



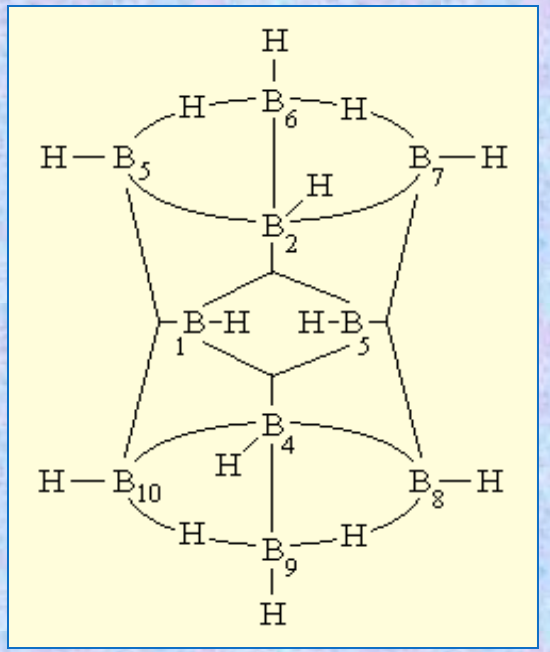
B_6H_{10}



B_5H_9

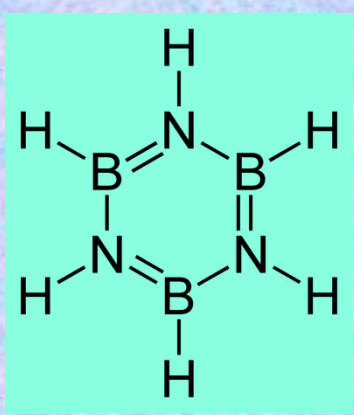


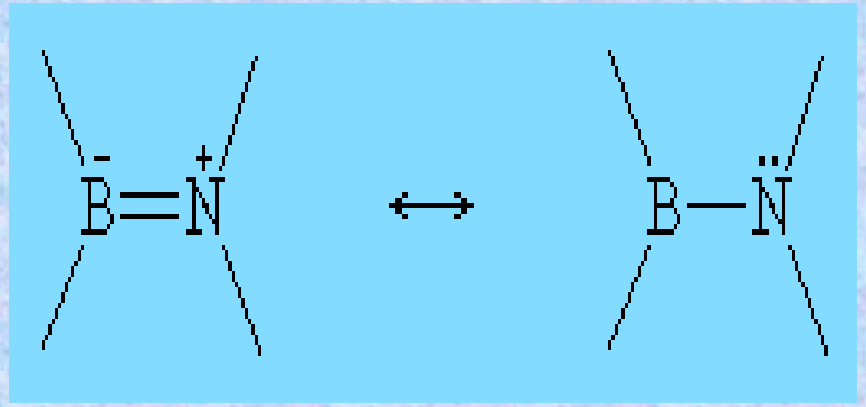
B_5H_{11}

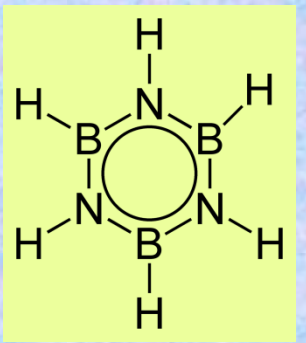


$B_{10}H_{14}$

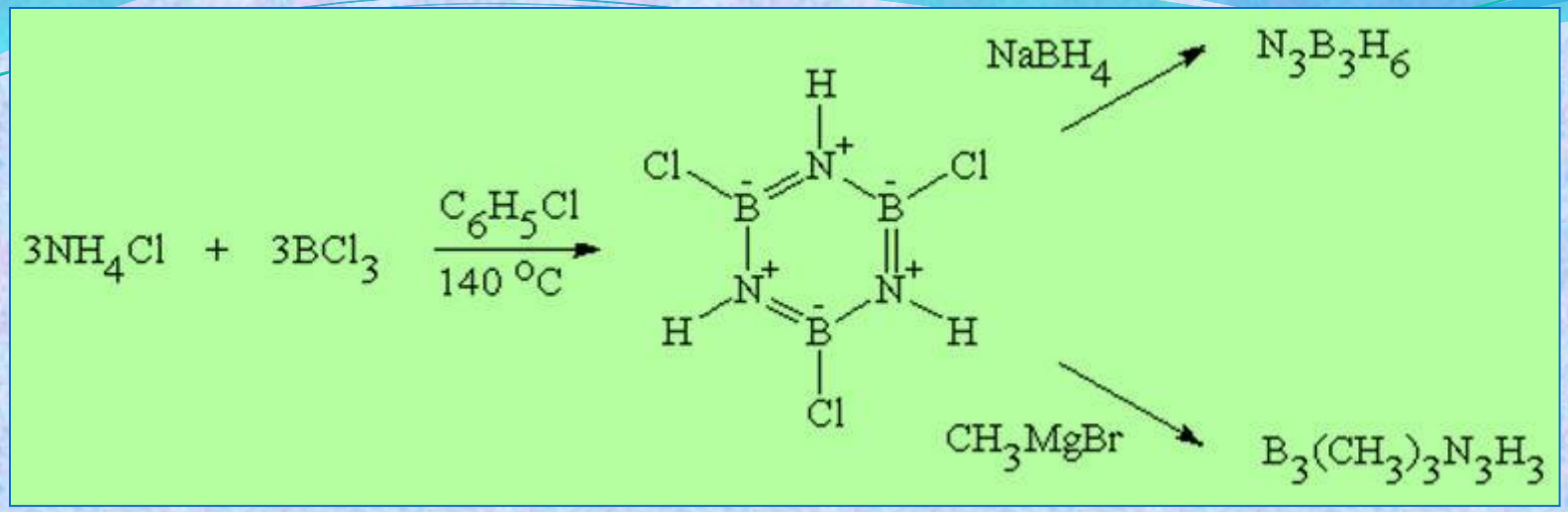
Borazine

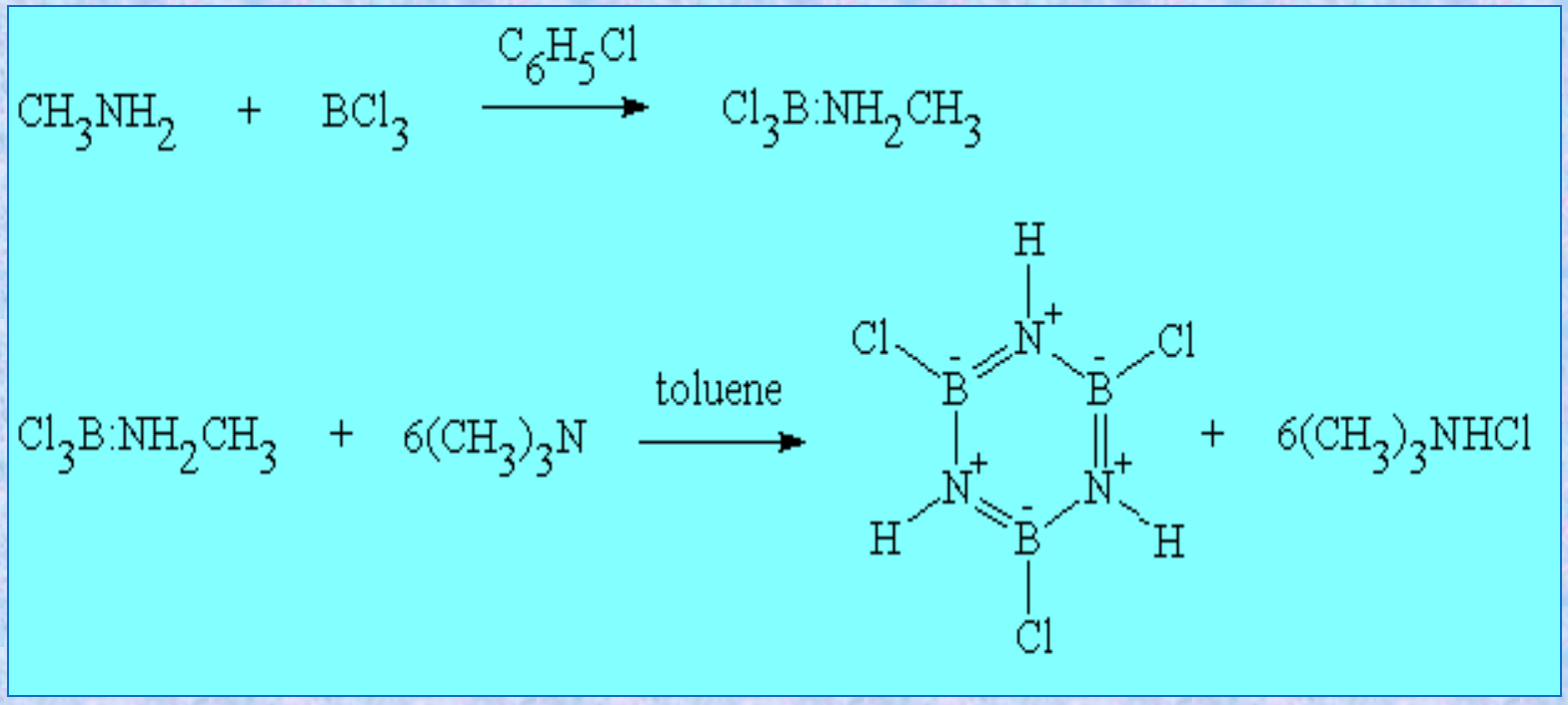




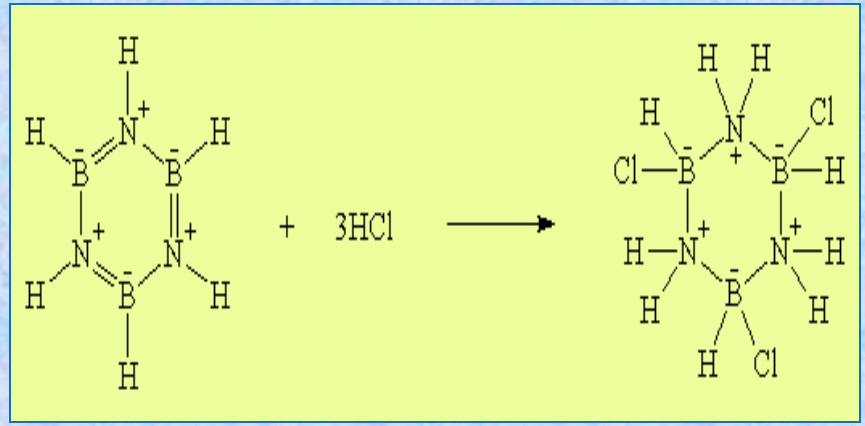


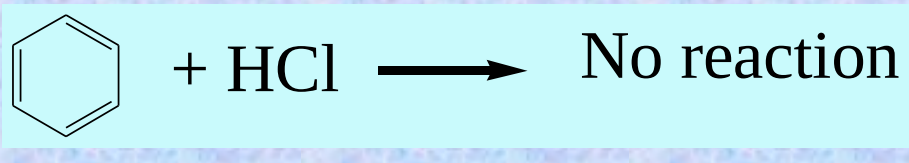
The molecules are flat, and rotation about the B—N bond is restricted, therefore the left-hand structure must be is a significant contributor. It is perhaps not correct to represent them as *canonical* structures since the geometries would be so different.


3NH4Cl + 3BCl3
C6H5Cl
140 °C
NaBH4
N3B3H6
CH3MgBr
B3(CH3)3N3H3

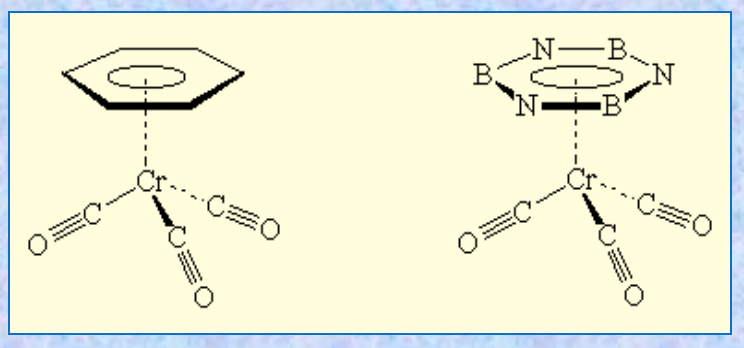

CH3NH2 + BCl3
C6H5Cl
Cl3B:NH2CH3
Cl3B:NH2CH3 + 6(CH3)3N
toluene
+ 6(CH3)3NHCl

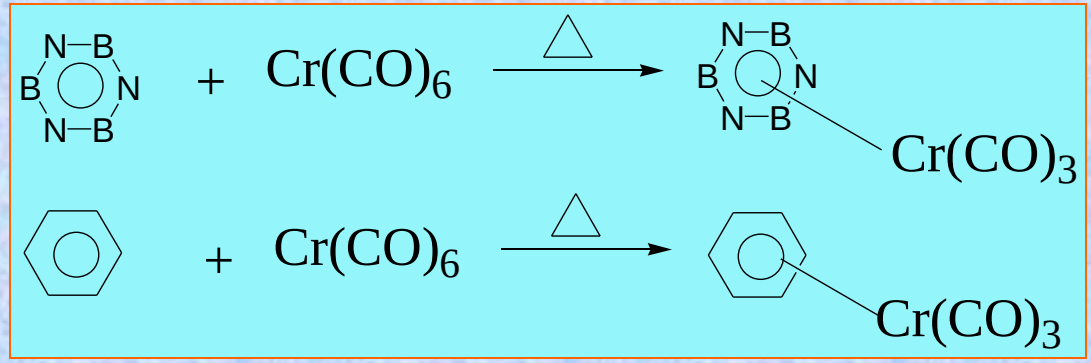
Unlike benzene, borazine undergoes addition reactions:





Like benzene, borazine can form π-complexes with transition metals:





- **What are the Similarities Between Borazine and Benzene?**

- **Both are cyclic structures.**
- **Also, both, borazine and benzene, have a six-membered ring.**
- **Moreover, both have equal bond lengths between the atoms of the ring.**
- **Furthermore, both are colourless liquids at room temperature.**
- **Both are isoelectronic as well.**

What is the Difference Between Borazine and Benzene?

- Borazine is an inorganic compound having the chemical formula $B_3H_6N_3$
- whereas Benzene is an organic compound having the chemical formula C_6H_6.
- Therefore, the key difference between borazine and benzene is that the borazine contains three boron atoms and three nitrogen atoms in the ring structure
- whereas benzene contains six carbon atoms in the ring structure. In borazine, the boron and nitrogen atoms are in an alternating pattern. As a result, the borazine molecule is not a perfect hexagon, but benzene has only carbon atoms in the ring. Thus, it can form a perfect hexagon. Hence, we can consider this as another difference between borazine and benzene.
- Above all, a significant difference between borazine and benzene is that the borazine is a synthetic compound whereas the benzene naturally occurs in crude oil as a major constituent. Also, we can identify a difference between borazine and benzene based on their reactivity as well. Comparatively, borazine is more reactive than benzene.

Comparsion of Borazine and Benzene

Definition	**Borazine is an inorganic compound having the chemical formula $B_3H_6N_3$**	**Benzene is an organic compound having the chemical formula C_6H_6**
Chemical Nature	**Inorganic**	**Organic**
Ring Structure	**Alternating Boron and Nitrogen atoms**	**Only carbon atoms**
Reactivity	**Comparatively more reactive**	**Comparatively less reactive**
Hexagonal Structure	**Not a perfect hexagon**	**A perfect hexagon**
Occurrence	**Synthetic**	**Naturally Occurs in Crude oil**

The Tetrahydroborate Ion (BH_4^-)

$NaBH_4$ is stable in dry air and alkaline aqueous solution. (It will react with water initially but the reaction stops as the concentration of the hydrolysis product, sodium borate, builds up.)

$LiBH_4$ is similar to $NaBH_4$ but more sensitive to water.

$Al(BH_4)_3$ is liquid which explodes with air or water. It probably has pairs of hydrogen bridges like diborane.

$Zr(BH_4)_4$ is a molecular solid with three bridging hydrogens connecting each boron to the zirconium.

Carbide

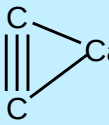
The binary compound of C with M is called carbide

Example : Al_4C_3, CaC_2

Types of Carbide :

1. **Ionic or Salt-like carbide :** C^{4-}, C_2^{2-}

$$Al_4C_3 + 12\ H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$$



$$+ 2\ H_2O \longrightarrow Ca(OH)_2 + H\text{-}C \equiv C\text{-}H$$

2. **Interstitial Carbide: Transition metal forms this carbide**
Structure: Closed-packed, Hard, Electric Conductor, High m.p.

WC, Fe3C, Mn3C, Ni3C- They are hydrolysed by Acids.

3. **Covalent Carbide: SiC, B_4C – Very Complex Structure, Very Hard,**

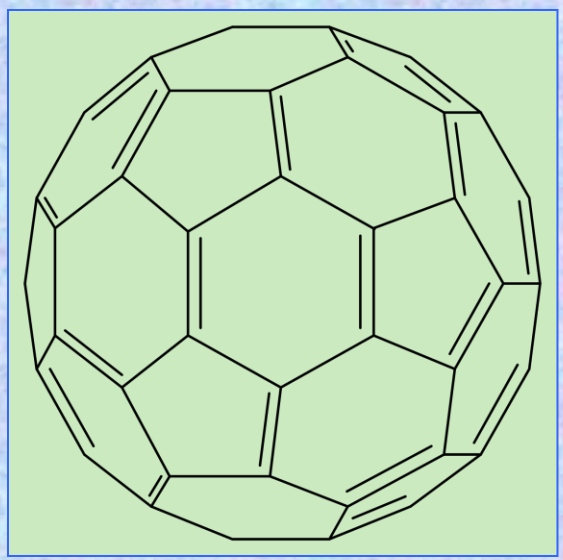
$$SiO_2(s) + C(s) \longrightarrow SiC(s) + 2CO(g)$$

$$SiC(s) \xrightarrow{\text{High Temp.}} Si(s) + C(s)$$

Graphite

Diamond Vs Graphite

Two crystal form of Carbon, Diamond and Graphite,
The another form of Carbon- Fullerene- C_{60}

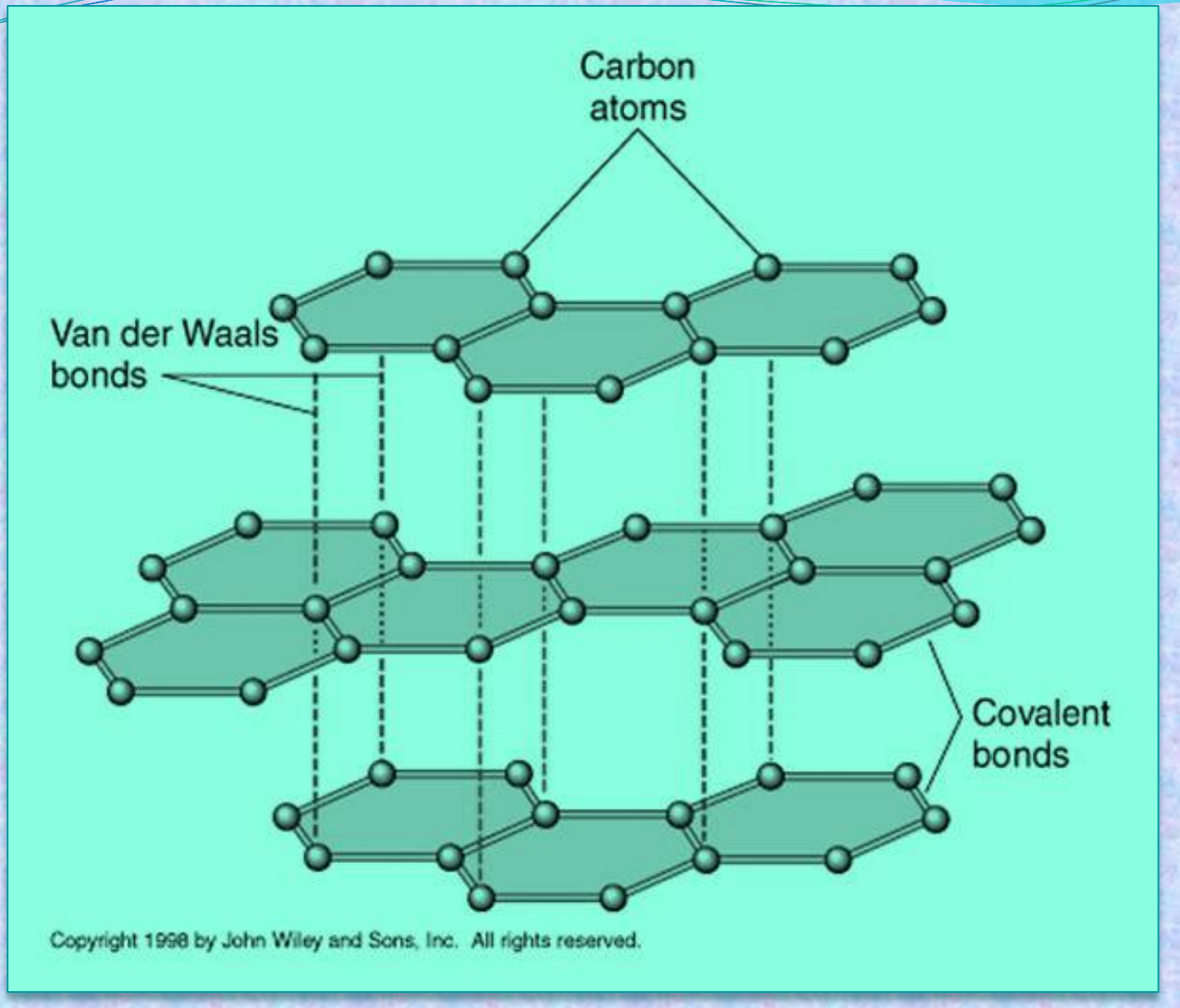


Diamond $\xrightarrow{1000\ ^oC}$ Graphite

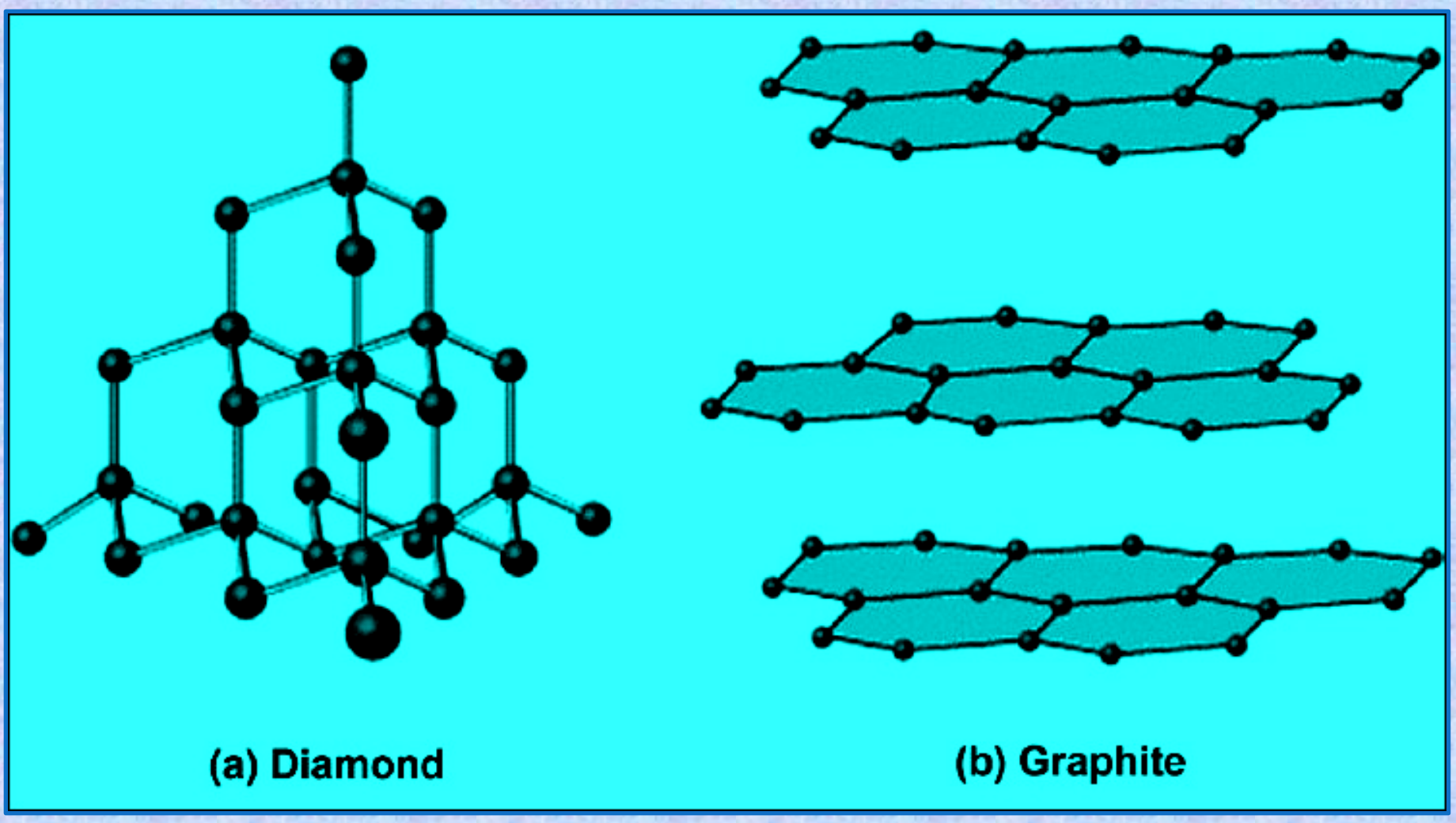
Graphite $\xrightarrow{2000^oC}$ Diamond

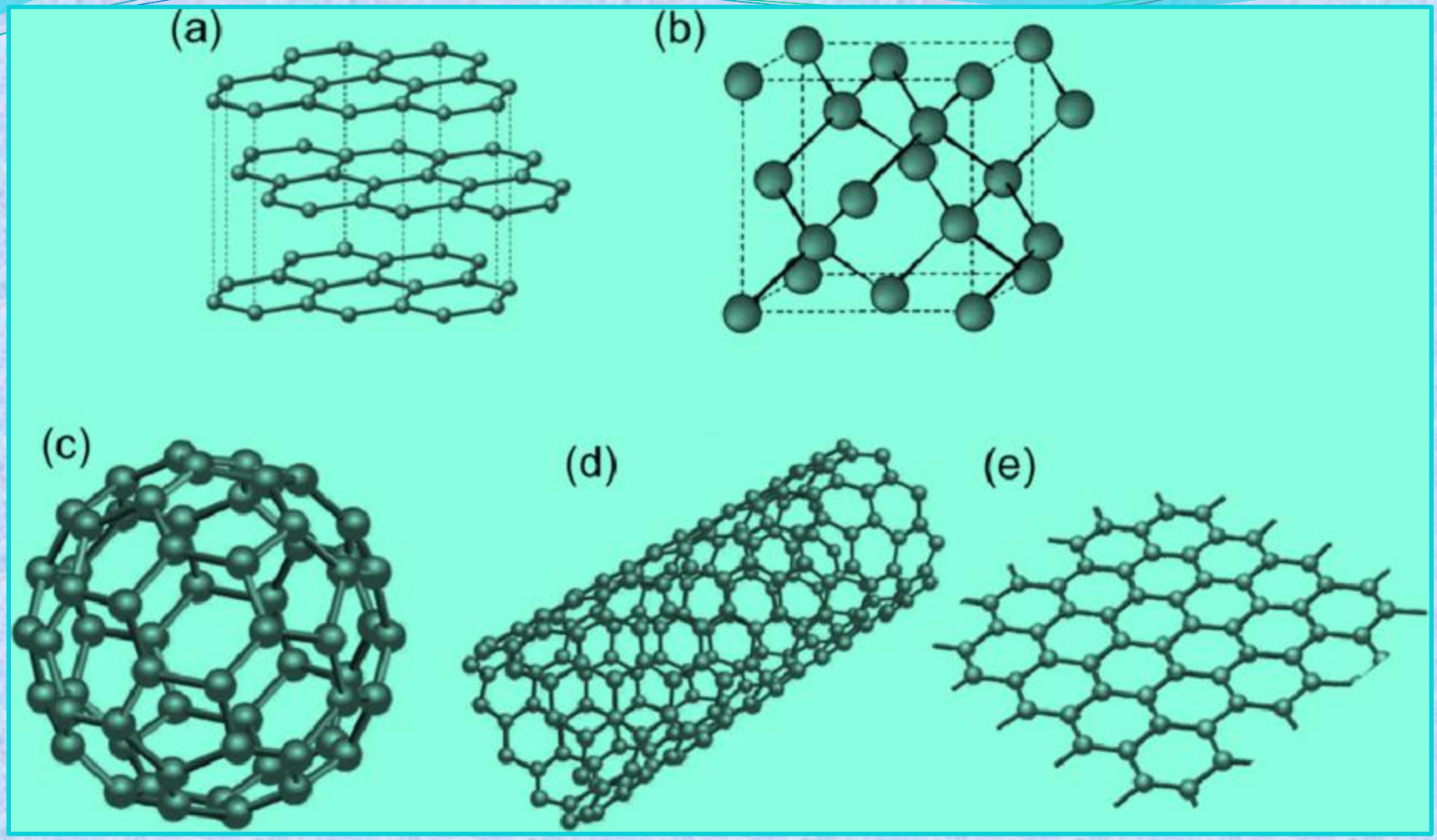
Comparison Between Diamond and Graphite

Diamond	Graphite
1. Tetrahedral Arrangement of C Angle 109°	1. Square Planer arrangement of C Angle- 120°
2. Three dimensional network	2. Layered structure (3.4 Å) separation
3. Sp^3 hybridization	3. Sp^2 hybridization
4.Non-conductor of electricity	4.Good conductor of electricity
5. No free electron	5. Free electron moves to a particular layer
6. Very hard	6. Soft and malleable
7. Density – 3.51 g/cm^3	7. Density – 2.22 g/cm^3



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(a) Diamond
(b) Graphite


(a)
(b)
(c)
(d)
(e)

THANK YOU